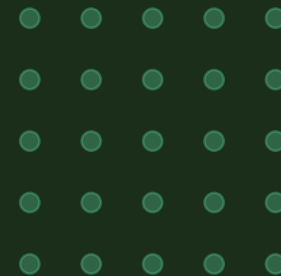




Data-Driven Policy to Reduce Food Waste in the United States



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*Core Finding: 3 causes × 3 sectors drive 71.5% of all preventable surplus.
Targeted federal policy prevents 116.9M tons · saves \$576B · recovers 194.7B meals.*

What We'll Cover

From the data to a winning federal policy

01**Why Food Waste?**

Scale of the crisis

02**Datasets & Sources**

ReFED + 4 live external APIs

03**EDA — National Trends**

Pareto, sectors, destinations

04**EDA — State Deep Dive**

Geographic targeting

05**ML Models (7 total)**

Classification, clustering, causal, anomaly

06**Causal Discovery**

Beyond correlation to causation

07**Counterfactual Sim.**

What-if policy levers

08**3-Pillar Framework**

Specific · Costed · Actionable

09**Impact & ROI**

\$576B, 194B meals, 292M mtCO₂e

10**Conclusions & Future**

The path to Wisconsin

A Crisis at the Intersection of Climate, Economy & Equity

1.0B

Tons wasted/yr
In the U.S. alone

\$4.5T

Economic loss
Annual food waste value

30%

Of U.S. GHGs
From food decomposition

1 in 8

Americans hungry
While food is discarded

Source: ReFED Food Waste Monitor (2010–2024)

Why does this problem need Data Science?

-  Root causes vary by sector — Farm, Foodservice, Residential each need a different solution
-  Geographic inequality — Washington state wastes 4× more (ratio) than Wisconsin
-  Value misalignment — by tons, farms are biggest; by dollars, restaurants dominate
-  No state is on track to meet the 2030 federal goal — data proves federal policy is essential

5 ReFED Files + 4 Live External APIs

Dataset	Rows	Columns	Coverage
cause_df — Cause Summary	16,455 rows	10 cols	2010–2024, 5 sectors, 9 cause groups
detail_df — Detail	13,560 rows	26 cols	sector × food type × destination
summary_df — Summary	4,350 rows	26 cols	aggregated across all dimensions
state_summary_df	216,930 rows	32 cols	50 states × 15 years
state_detail_df	646,137 rows	31 cols	granular state-level breakdown

External Datasets



USDA ERS Food Insecurity

765 rows · 51 states × 15 years ·
food_insecurity_pct



NOAA NCEI Temperature

750 rows · 50 states × 15 years ·
avg_temp_f (live API)



USDA NASS Produce Value

765 rows · 36 commodities ·
produce_value_per_capita



US Census Population

728 rows · 50 states × 14 years ·
population data

Clean Data → Engineered Features → Master Panel



Data Quality Check

0 missing values across all 5 ReFED files
Descriptive stats validated (max tons = 5.8M, max value = \$61.8B)

Feature Engineering

Derived ratios: surplus_ratio, recovery_rate, landfill_rate, harvest_loss_rate, ghg_per_ton, value_per_ton

Master Merge

ReFED + ERS + NOAA + NASS + Census
Join key: (year, state) — left join, 28 final columns

Master State (ML ready)

Averaged 2019–2024
StandardScaler applied before clustering & regression

NOAA Deduplication

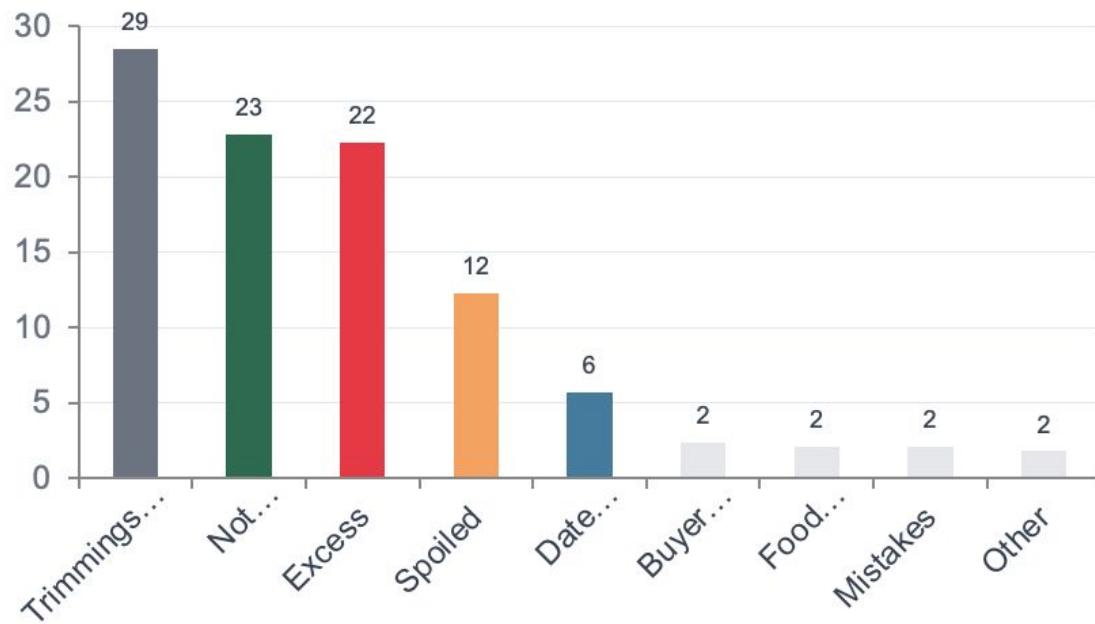
Temperature: 50 states × 15 years — $r=+0.183$
with surplus ratio (warmer → more spoilage)

NASS Deduplication

11,550 → 1,363 rows (removed sub-category duplicates)
Only 'ALL UTILIZATION PRACTICES' + 'ALL CLASSES' kept

3 Cause Groups Drive 71.5% of All Preventable Surplus

By Weight — % of Total Tons



1,017M

Total tons

2010–2024 cumulative

727.7M

Preventable

71.5% — the policy universe

\$2.3T

Excess value

50.6% of all economic value

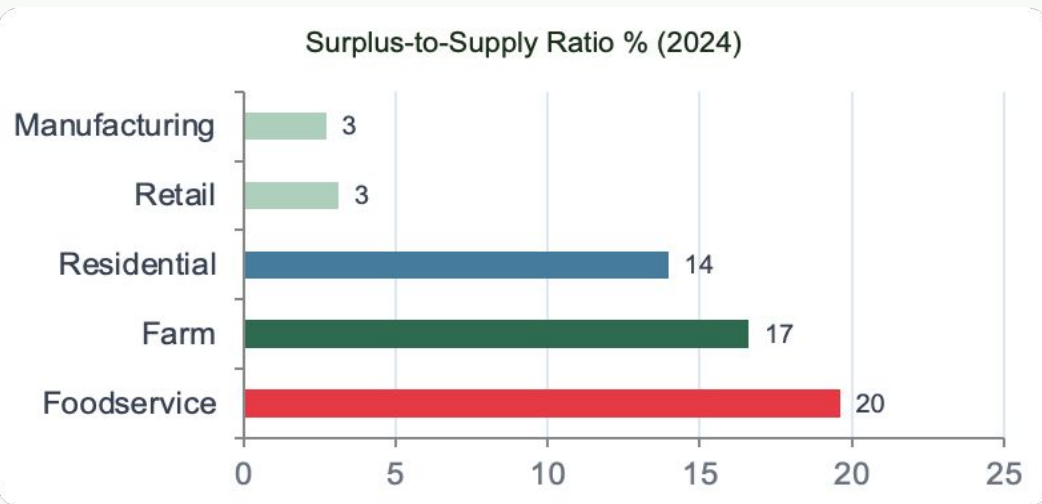
119.8M

Plate Waste

Single largest specific cause

KEY: Tons → Farm policy · Dollars → Foodservice policy

Foodservice Wastes 20% of Its Supply Every Year



💡 COVID-19 Natural Experiment — Proof of Concept

2020: ratio fell –25.8% when restaurants closed. Recovery by 2024 was +26.4% above the 2020 low. This proves rapid, policy-driven reduction is achievable — not just theoretical. If closures caused it, targeted mandates can replicate it.

National Surplus Ratio 2010–2024

2010	0.0801	
2014	0.0760	Best pre-COVID
2019	0.0778	Pre-COVID baseline
2020	0.0577	★ –25.8% COVID
2021	0.0640	Partial recovery
2022	0.0721	+12.7%
2024	0.0730	Slight improvement

Each Cell = One Policy Target

Sector	Not Harvested	Excess	Spoiled	Date Labels	Buyer Rej.	TOP TARGETS
Farm	231.6 ★	5.0	30.5	0.0	9.8	P1
Foodservice	0.0	153.5 ★	3.5	8.2	0.0	P2
Residential	0.0	59.7	78.8 ★	18.1	0.0	P3
Retail	0.0	4.9	12.6	31.7 ★	0.0	P3
Manufacturing	0.0	4.2	0.0	0.0	14.6	

★ 3 cells drive the entire 3-pillar framework: Farm×Not Harvested (231.6M) → Pillar 1 · Foodservice×Excess (153.5M) → Pillar 2 · Retail×Date Labels (31.7M) → Pillar 3

Plate Waste (119.8M tons) = The Biggest Specific Cause in the Dataset

Top Preventable Specific Causes (Million Tons)

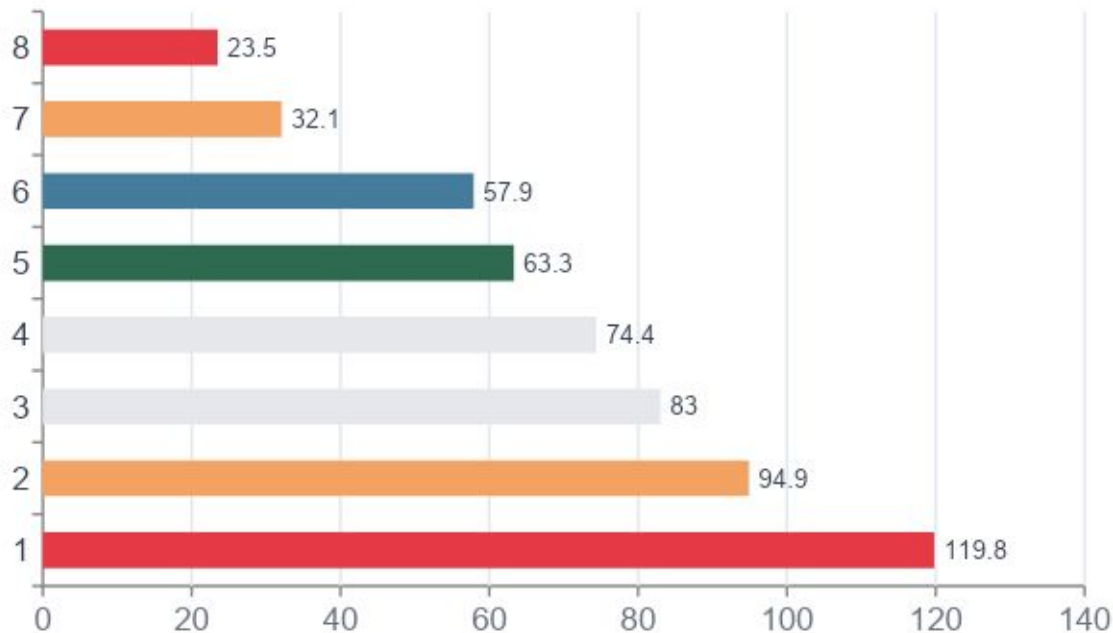


Plate Waste 119.8M

Portion transparency + consumer nudges + take-home incentives

Left Behind Marketable 63.3M

Federal gleaning mandate — already piloted CA/NY → Pillar 1

Date Labels 57.9M

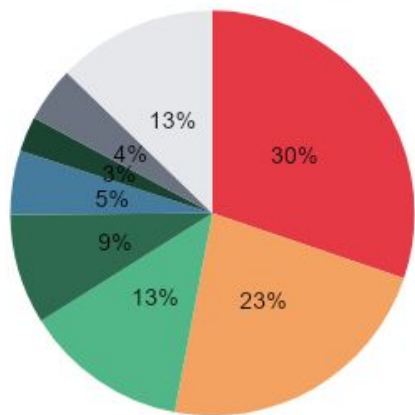
ONE federal rule: standardize to 2 labels only → Pillar 3

Overproduction 23.5M

AI demand forecasting mandate for 50+ location chains

Only 33.6% Recovered - 38.5% Goes Straight to Landfill

Where 1,017M Tons End Up (2010–2024)



- Landfill (worst)
- Not Harvested
- Animal Feed
- Composting
- Anaerobic Dig.
- Donations
- Incineration
- Other recovery

341.7M

Tons recovered

Donations + feed + composting + AD

391.8M

Tons truly wasted

Landfill + incineration + sewer

1,652B

Meals wasted

2010–2024 cumulative

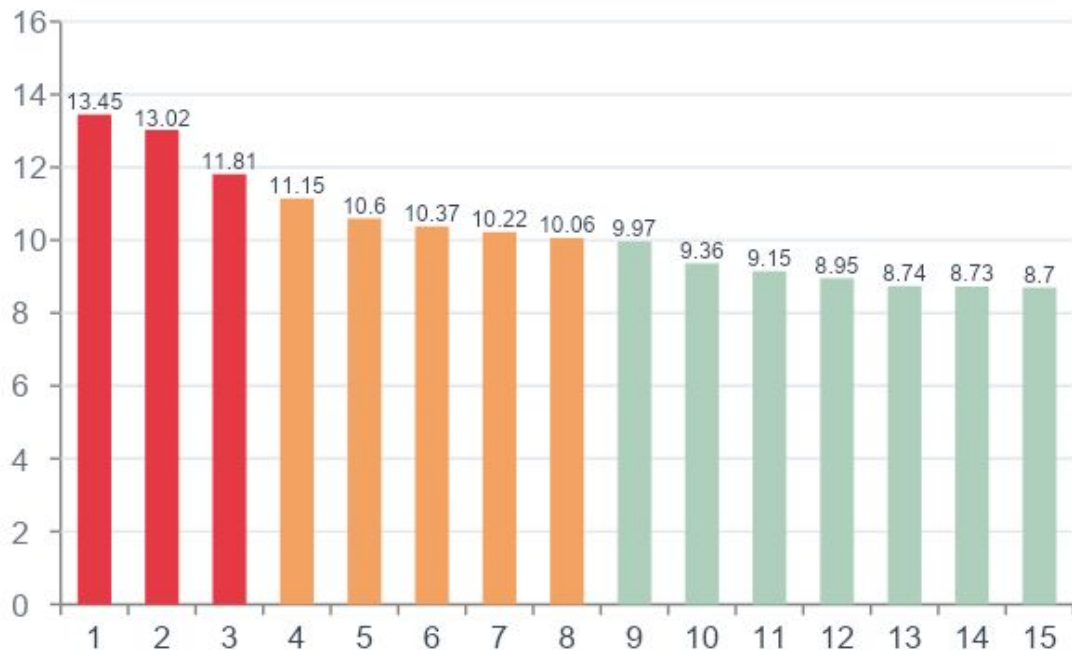
3.02B

mtCO₂e total GHG

Upstream + downstream

Washington Wastes 4× More Than Wisconsin — Geography Matters

Top 15 States — Surplus-to-Supply Ratio % (2024)



State Priority Matrix

⚡ URGENT (2 states)

MA + MT: High ratio + Worsening trend

🟡 High + Improving (23)

WA,AZ,NV,HI,WV,NH,CA,WY,MD,SC...

🟢 Monitor (25)

Most Midwest states

Best Performers (2024):

Wisconsin: 3.19% — Lowest nationally

Louisiana: 3.76% — 61.9% recovery

N.Dakota: 3.96% — 64.3% recovery

12 States Waste Food Where People Go Hungry

HIGH WASTE + HIGH INSECURITY ★

AZ, CA, FL, GA, MI, NV, SC, TX...

12 states — strongest case for federal intervention

High Waste, Low Insecurity

CO, CT, HI, ME, MD, MA, NH, WA...

Economic waste — less equity urgency

Low Waste, High Insecurity

AL, AR, IL, IN, KY, MS, OK, TX...

Recovery gap — food not reaching communities

Low Waste, Low Insecurity ✓

DE, ID, IA, KS, MN, NE, ND, WI...

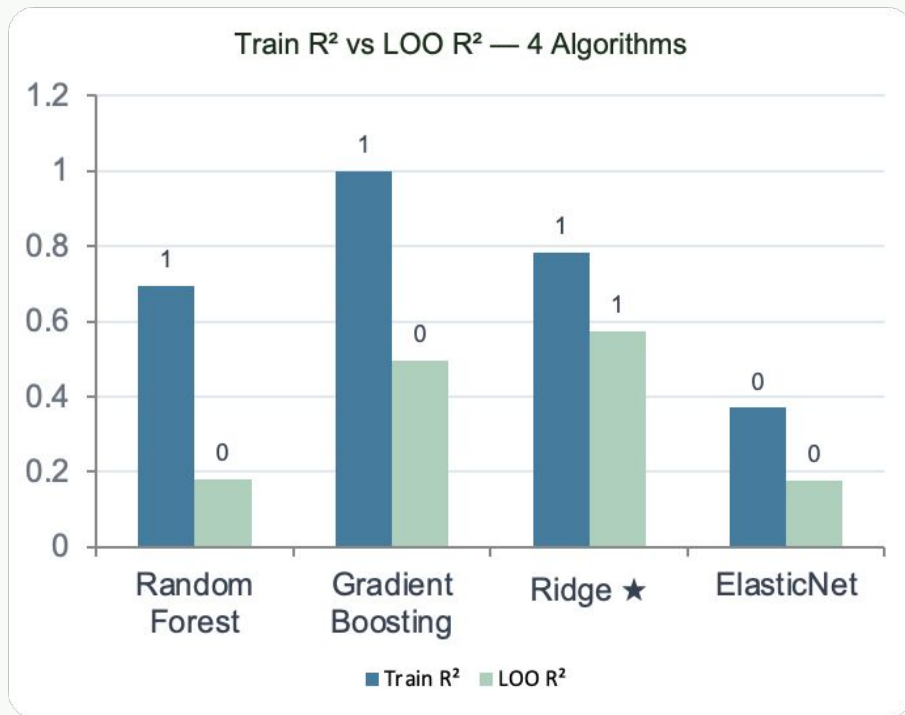
Model performers — study and replicate

← LOW FOOD WASTE HIGH FOOD WASTE →

From Prediction to Causation to Prescription

M1 4-Algorithm Comparison	Ridge LOO $R^2=0.574$ RF · GB · Ridge · ElasticNet LOO CV on 50 states × 9 features	M2 Cause-Level GradientBoosting	Test $R^2=0.885$ n=16,455 · 80/20 stratified split Validates 3-pillar targeting	M3 Sector Clustering	Silhouette=0.944 K-Means k=5 on sector×year profiles 5 distinct policy archetypes	M4 State Clustering	Ward k=3, sil=0.490 3 archetypes: High Landfill · High Recovery · High Ratio
M5 State Trajectory & Forecast	0/50 states on track No state meets 2030 goal alone BAU: 63M tons · Policy: 51M tons	M6 Causal Discovery	PC Algorithm harvest_loss → surplus_ratio (causal!) Beyond correlation to root causes	M7 Counterfactual + Anomaly	4.8M tons/yr #1 lever What-if simulations + Isolation Forest Wisconsin: 57% below predictions		

Ridge Best (LOO $R^2=0.574$) — RF & GB Agree on Top Features

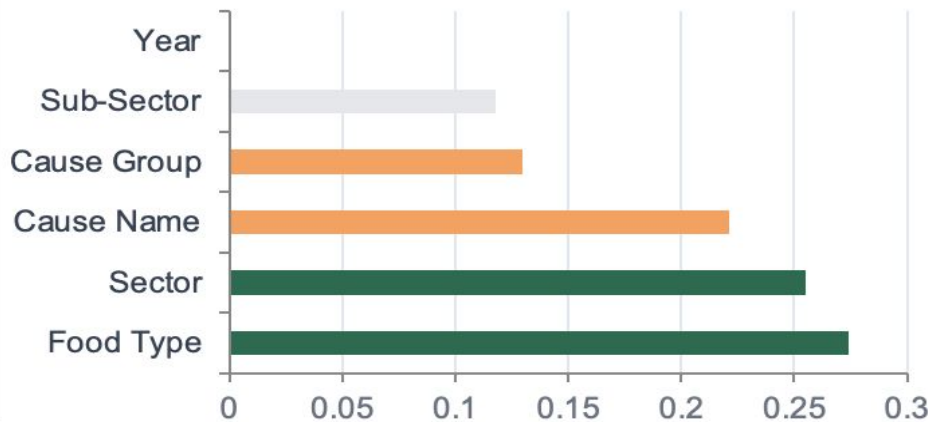


Feature	RF	GB	Policy Link
harvest_loss_rate	0.210	0.412	→ Pillar 1 gleaning ★
value_per_ton	0.203	0.298	→ Pillar 2 foodservice ★
recovery_rate	0.119	0.074	→ All pillars diversion
landfill_rate	0.108	0.054	→ Waste destination KPI
ghg_per_ton	0.093	0.077	→ Environmental driver
avg_temp_f	0.077	0.016	→ Cold-chain need
food_insecurity_pct	0.074	0.033	→ Equity signal
trend_slope_ratio	0.068	0.012	→ Urgency indicator
produce_value/cap	0.047	0.024	→ Pillar 1 targeting

★ Both RF and GB rank harvest_loss + value/ton #1 and #2 — model-independent validation

Test $R^2=0.885$ on 16,455 Rows — ML Independently Validates 3 Pillars

Feature Importance — Cause Model



0.895

Train R^2

0.885

Test R^2 ★

13,164

Train rows

3,291

Test rows

Sector × Cause	Actual	Pred	Err
Farm × Not Harvested	231.6M	205.5M	11% ✓
Foodservice × Excess	153.5M	78.1M	49% ⚠
Residential × Spoiled	78.8M	63.7M	19% ✓
Residential × Excess	59.7M	49.7M	17% ✓
Retail × Date Labels	31.7M	18.7M	41% ⚠
Farm × Spoiled	30.5M	11.8M	61% ⚠
Model confirms Farm×Farm, Residential×Spoiled, Excess — validating our 3 pillars			

0 Out of 50 States Are On Track to Meet the 2030 Goal

USDA/EPA Target: 50% reduction from 2016 baseline by 2030

0

out of 50 states

on track for
2030 goal

★ **STRONGEST ARGUMENT FOR
FEDERAL POLICY**

0

On Track

Zero states—federal action essential

1

Close — Needs Acceleration

Florida (close but needs help)

47

Will Miss Target

AL AK AZ AR CA CO CT DE GA HI ID IL IN IA KS KY LA ME MD MI MN MS MO NE NV NH NJ NM NY NC ND OH
OK OR PA RI SC SD TN TX UT VT VA WA WV WI WY

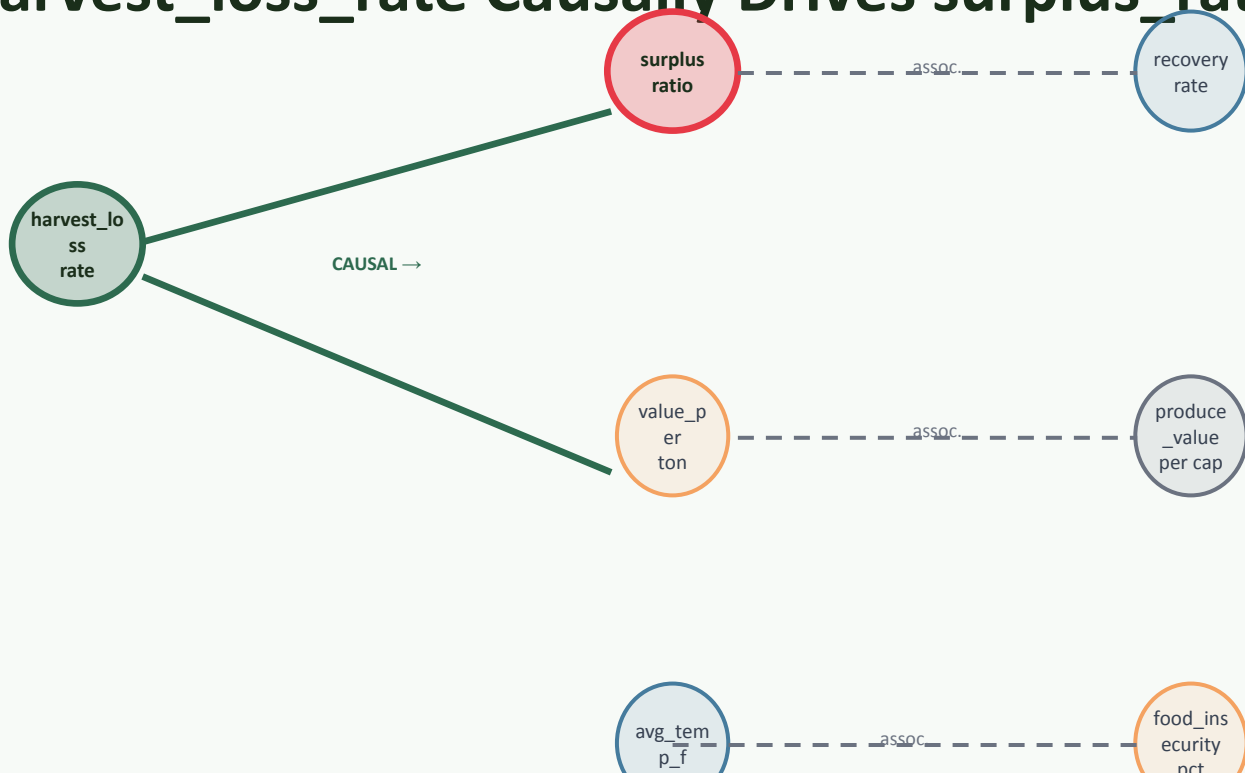
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Moving WRONG Direction

Massachusetts, Montana (worsening)

States can't get there alone. Federal policy is the only path.

harvest_loss_rate Causally Drives surplus_ratio



Policy Implication

harvest_loss
→ surplus

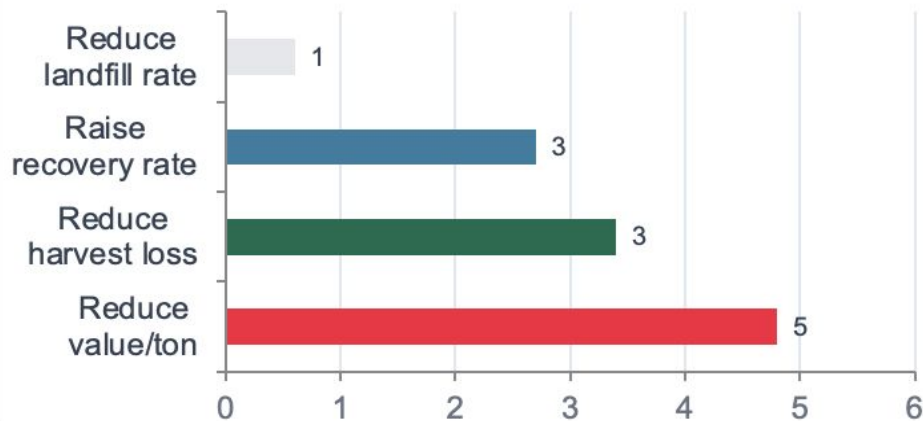
ROOT CAUSE
intervention

Pillar 1 is
the highest-
leverage
policy lever

This upgrades SHAP from 'correlated with' to 'causally drives' — Pillar 1 targets a root cause, not just an association

What-If Policy Levers + Which States Defy Expectations

Counterfactual: If All States Reach Top Quartile
(Tons Prevented/Year)



#1 lever: Reduce value/ton (4.8M tons/yr) → Pillar 2 portion standards + demand forecasting

#2 lever: Reduce harvest loss (3.4M tons/yr) → Pillar 1 gleaning mandates

Anomaly Detection — Isolation Forest

States wasting MORE than predicted:

Washington	13.0%	7.2%	+5.7%
Arizona	12.2%	7.6%	+4.6%
Nevada	10.8%	7.6%	+3.2%
W.Virginia	9.8%	7.0%	+2.8%

States wasting LESS than predicted (policy models):

Louisiana	3.8%	6.5%	-2.9%
Wisconsin	3.1%	7.4%	-4.3%
N.Dakota	4.1%	6.5%	-2.5%

Wisconsin Wastes 57% Less Than Predicted — Here's Why



Wisconsin

Actual ratio:

3.1%

Predicted by model:

7.4%

57% below prediction
→ unexplained by data alone

If ALL states matched Wisconsin:
70M → 30M tons (-57%)

Why Wisconsin Outperforms — State Policies to Replicate Federally

1

Wisconsin Act 287 (2013)

Required large food generators to divert organics from landfill — one of the first mandatory organics diversion laws in the US → Federal equivalent would help all 50 states

2

Clean Sweep Program

State-funded food recovery grants to food banks and composting facilities → Federal matching grants program

3

UW-Extension Research

Research and extension services to farms and processors → Federal ag extension investment

4

Dairy Culture

Our 3-pillar policy is the realistic federal path to Wisconsin-level performance nationally

Specific · Costed · Evidence-Backed · Actionable

Each pillar targets the highest-impact cause×sector combination validated by ML

1

**Farm:
Not Harvested Produce**

46.3M tons

\$30B

tons prevented

value saved

- Federal gleaning mandate (farms >500 acres)
- Crop insurance: cover price-collapse losses
- USDA Produce Forward Contract Program
- Emergency harvest tax credit
- Quick win: already piloted CA + NY

2

**Foodservice:
Excess Overproduction**

30.7M tons

\$356B

tons prevented

value saved

- Mandatory surplus reporting (50+ locations)
- Tax credit: AI demand forecasting adoption
- Federal portion-transparency standard
- Good Samaritan Act expansion
- Quick win: liability clarification for apps

3

**Retail + Residential:
Date Label Reform**

10.0M tons

\$45B

tons prevented

value saved

- 2 labels only: 'Best If Used By' + 'Use By'
- Retailer clearance incentive (tax deduction)
- Consumer nudge: smart fridge app standard
- K-12 food storage literacy curriculum
- Quick win: bill drafted (USDA/FDA 2019)

20% Reduction Across 3 Pillars = Transformative Impact

116.9M

Tons prevented

20% × 584.3M baseline

\$576B

Value saved

Cumulative economic

194.7B

Meals recovered

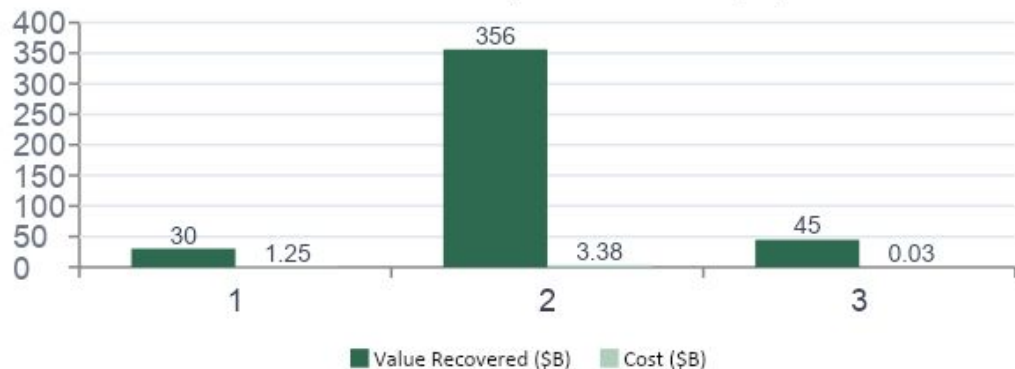
1,666 meals/ton (NRDC)

292M

mtCO₂e avoided

2.5 mtCO₂e/ton (EPA)

Value Recovered vs Implementation Cost (\$B)



★ Pillar 3 = highest-ROI public policy intervention in the ReFED dataset

24×

ROI

P1 Gleaning

\$1.25B → \$30B

106×

ROI

P2 Foodservice

\$3.38B → \$356B

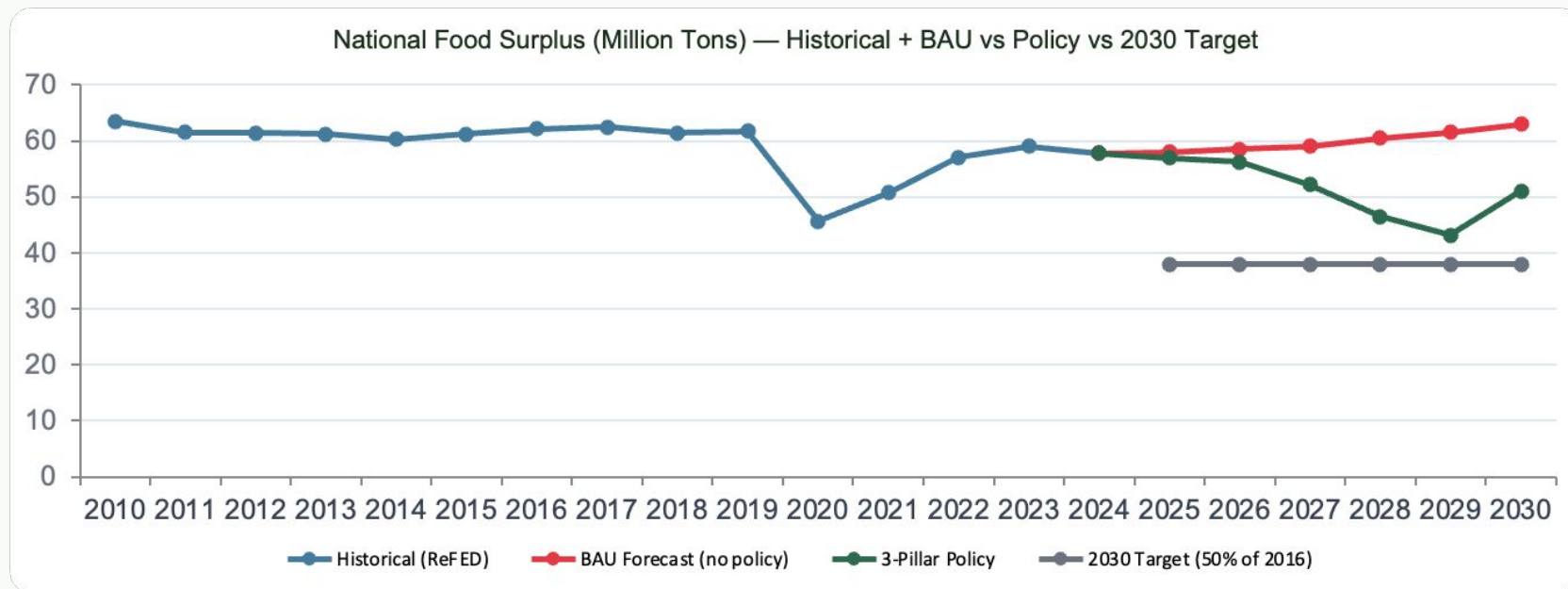
~1,493×

ROI

P3 Date Labels

\$30M → \$45B

BAU Misses 2030 Goal by 25M Tons — Policy Cuts the Gap in Half



COVID –25.8%

Rapid reduction IS achievable

BAU misses by 25M tons

Trend alone is not enough

Policy: 13M tons saved

Closes half the gap by 2030

7 Data-Backed Findings That Demand Federal Action

01

Waste is concentrated

3 causes × 3 sectors = 71.5% of preventable surplus — proven from 16,455 rows

03

Only 33.6% recovered

38.5% to landfill. Methane from landfill is 80× more potent than CO₂ over 20 years

05

Harvest loss is causal

PC algorithm confirms harvest_loss_rate → surplus_ratio. Pillar 1 targets a root cause

07

Wisconsin shows the way

57% below predictions due to state policy — at national scale: 70M → 30M tons

02

Plate Waste leads

119.8M tons — single largest specific cause, bigger than ALL Not Harvested sub-causes

04

COVID proved it works

−25.8% in one year. Policy can replicate this deliberately through mandates

06

0/50 states on track

No state meets 2030 goal on current trajectory. Federal action is the only path

Beyond the Datathon — 6 Paths to Greater Impact



AI Demand Forecasting at Scale

Mandate AI/ML inventory systems for chains with 50+ locations. Pilot with DoorDash/Grubhub APIs. Estimate: prevents 3–5M additional tons/year.



Real-Time National Dashboard

Build a live ReFED × NOAA × USDA dashboard tracking progress toward 2030 goal per state. Publicly available — creates accountability.



Expand Causal Analysis

Larger n (county-level) enables stronger causal discovery. Add poverty rates, cold-chain infrastructure density as new variables.



Natural Language Processing

Mine USDA inspection reports + local news for food waste events. Adds qualitative policy context to quantitative ReFED data.



County-Level Targeting

Disaggregate state data to county level for surgical policy targeting. 50 states → 3,143 counties = 63× more precise intervention.



Economic Policy Modelling

Microsimulate the Produce Forward Contract Program impact on farm-level price stability using USDA NASS price data.



Three Laws. One Decade. A 69% Reduction in Food Waste.

1

The FARM Gleaning Act

Require farms >500 acres to allow certified gleaning. Cost: \$1.25B. ROI: 24x. Precedent: CA + NY already piloted.

2

Foodservice Transparency Rule

Surplus reporting for 50+ location chains. AI forecasting tax credit. Good Samaritan expansion.

3

Date Label Standardization Act

Mandate 2 labels only. Cost: \$30M. ROI: ~1,500x. Bill drafted 2019 — needs political will.